

Flavio Smirne

PRINCIPAL DATA ARCHITECT

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PROFESSIONAL SUMMARY

Hands-on data and platform architect with 20+ years designing enterprise data platforms, integration frameworks, real-time decisioning systems, and large-scale analytics architectures across retail, financial services, e-commerce, and insurance. Sets data architecture direction across entire product portfolios — author of canonical enterprise data models, semantic and governance frameworks, and re-platforming strategies adopted by 60+ of the top 100 U.S. retailers and underpinning ~40% of U.S. omnichannel transactions. Pairs deep data modeling and system design with the cross-organizational influence to align engineering, data science, product, and executives on a single technical strategy — owning the design decisions and trade-offs end to end.

CORE TECHNICAL EXPERTISE

Architecture: Enterprise Data Architecture, Solution & Integration Architecture, Event-Driven & API Architecture, Data Modeling & Information Architecture, Platform Modernization & Re-Platforming

Data & Platforms: Snowflake, dbt, BigQuery, Azure Data Platforms, PostgreSQL, SQL Server, ETL/ELT Pipeline Design, Data Quality Frameworks, Self-Service Analytics

AI/ML & Decisioning: AI/ML Enablement (LLMs, traditional ML), Real-Time Decisioning & Scoring, Fraud Decisioning Platforms, Risk Scoring, AML/KYC, Identity & Account Protection

Languages & Tools: Python, C#/.NET, SQL, JavaScript, REST APIs, Tableau / Power BI / Splunk

Governance & Practices: Data Governance & Standards, Architecture Review & Standards-Setting, Cross-Functional Technical Leadership, Vendor & Technology Evaluation

WORK EXPERIENCE

Senior Director of Implementation — Appriss Retail, Irvine, CA (Remote) 12/2024 – Present

- Own the enterprise technical architecture and data strategy across the full product portfolio — POS and e-commerce data spanning over \$1 trillion in annual transactions and ~250M unique customer identifiers, representing ~40% of U.S. omnichannel transactions across 60+ of the top 100 U.S. retailers.
- Architected the Appriss Retail Data Model v3, the canonical enterprise data model serving as the single foundation for analytics, fraud detection, and AI/ML return-authorization decisioning across every product.
- Personally wrote the first POS and e-commerce parsers on ARDM v3 as a working proof-of-concept, validated against a live client across two delivery phases (POS, then e-commerce) — establishing the reference implementation that anchors the new ingestion model.
- Established enterprise data standards, semantic models, and governance frameworks that drive consistency and interoperability across product domains and client integrations — authoring core framework code that standardizes and validates ARDM v3 across all clients.
- Leading the re-architecture of the retail ingestion platform from Azure/Python pipelines feeding a Greenplum MPP cluster to a Snowflake/dbt ELT ecosystem — adopting Snowflake's separation of storage and compute to eliminate resource contention between ingestion load and analytics and to simplify elastic scaling across all clients.
- Built an LLM-backed data discovery tool (Claude) that auto-generates schema analysis of raw client data and produces a working conversion prototype mapping it to ARDM v3 — cutting client implementation timelines by 40%.
- Serve as principal technical advisor to engineering, product, customer success, and executive leadership on data strategy and platform architecture.

Director, E-Commerce Fraud — GameStop, Grapevine, TX (Remote)

03/2022 – 12/2024

- Re-architected GameStop's global e-commerce fraud platform around an ML-first decisioning model, relegating the legacy rules engine to a thin policy layer and cutting the maintained rule set by two-thirds.
- Reduced the fraud rate by 68% by shifting the bulk of decisioning from hand-tuned rules to ML models, with real-time scoring and data-driven policy optimization.
- Designed real-time fraud decisioning systems integrating ML models, third-party data providers, behavioral analytics, and risk signals into a unified scoring pipeline.
- Architected the fraud analytics and reporting platform on GCP BigQuery, Amazon S3, and REST integrations, enabling data-driven decisioning across fraud, finance, and executive leadership.
- Personally wrote much of the team's .NET tooling — including a bulk fraud-resolution client integrated directly with Accertify and the GameStop Dispute Management System, which generated specialized financial and analyst-performance reports and produced every fraud and chargeback report delivered to the C-suite.
- Designed fraud-intelligence data models for account-takeover detection, card-testing prevention, AML monitoring, chargeback analytics, and transaction-fraud prevention.

Solution Productization Architect — Accertify (American Express), Itasca, IL 01/2014 – 03/2022

- Designed the airline data model from the ground up — the foundational schema behind fraud systems serving 8 of the 10 largest global airlines (United, Delta, Southwest, Air France-KLM) — and defined the retail and ticketing vertical models deployed to every Accertify client globally as the basis for fraud detection, analytics, and reporting.
- Architected platforms that sustained peak loads of 50M+ transactions and over \$6B in processed volume on a single Black Friday.
- Re-engineered core fraud detection algorithms, materially improving accuracy and reducing false positives.
- Architected self-service analytics platforms that became a key differentiator in client retention and new-business acquisition.
- Delivered highly available enterprise fraud platforms with zero downtime, coordinating across engineering, data science, product, and client services.
- Recipient of the American Express President's Award for top 1% global performance.

Lead Developer, Data & Mobile Platforms — State Farm Insurance, Bloomington, IL 04/2008 – 01/2014

- Designed the data aggregation system that served as the central orchestrator for all data requests across mobile and web applications — a unified API layer consolidating data services from disparate enterprise backends.
- Led development of State Farm Pocket Agent for Windows Phone, one of the industry's earliest mobile insurance applications.

CTO — Realinked.com, Chicago, IL 02/2010 – 08/2011

- Designed the company's end-to-end technology architecture from zero — data platform, analytics infrastructure, and integration framework — scaling it through profitability to a successful exit.

Software Engineering Manager — Wolfram Research, Champaign, IL 03/2004 – 04/2008

- Managed the Import/Export team, owning every data format supported in Mathematica — dozens of file formats across data ingestion and export.
- Personally wrote many of the file importers and authored the Rasterize function that converts Mathematica expressions into raster images — still core to the product today.

EDUCATION

Bachelor of Science, Computer Engineering — University of Illinois, Champaign-Urbana (<i>Dean's List</i>)	2003
Bachelor of Science, Civil Engineering — University of São Paulo, São Carlos, SP, Brazil	1999

LANGUAGES

English | Portuguese | Spanish